SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded

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CO2
Assesment Tool - 1

① Compare type 1 and type 2 Hypervisor for Hosting Mission.

i) Introduction:

A hypervisor is a software that creates and manages virtual machines on a physical computer. Each virtual can run its own operating system and applications.

Hypervisor are mainly classified into two, type-1 : type-2. They help cloud data centre use hardware efficiently by running multiple virtual machine on a single physical server.

i) Type 1:
Hypervisor (Bare - Metal Hypervisor)

A type - 1 Hypervisor runs directly on the physical hardware without requiring a seperate host operating system. it manages the CPU, memory, storage and network resources allocated to virtual.

ii) Type 2:
Hypervisor (Hosted Hypervisor)

A type - 2 Hypervisor runs as an application on top of an existing operating system, such as windows or linux. The host operating system manages the Hardware while the Hypervisor runs virtual Hypervisor.

(2) Workload isolation in a cloud Environment.

Introduction:

Workload isolation is the process of seperating application to prevent interference and unauthorized access. Virtualization allows a cloud provider to run finance, health care and student portal application on the same physical servers using seperated virtual machines

Role of Virtualisation:

- Finance: protects financials transaction and customer data
- Health care: Ensure the confidentialites of patient record

Two major attach surface:

1. Hypervisor: Attackers may exploit Hypervisor weakness to Access other VMs

2. Virtual network vulnerabilities :- incorrect network configuration may allow communication between VMs.

Conclusion:

Virtualization provide workload isolation through seperate VM and controlled resources allocation. A secure Hypervisor, proper network configuration and access control help protect sensitive data and ensure reliable cloud service

(3) VM utilization Analysis in Cloud computing

VM utilization Analysis is a process of measuring and evaluating how efficiently virtual machine use resource such as CPU, memory, storage, in a cloud environment

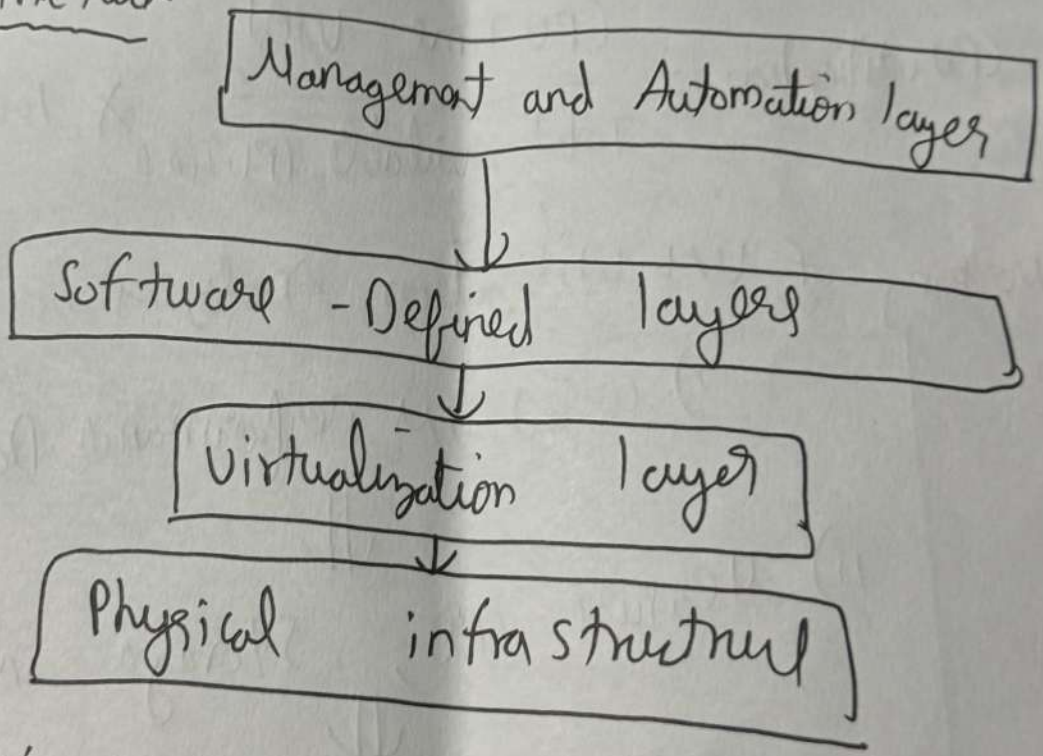
Important Parameters:

: Percentage of CPU capacity used by a virtual Machine

④ Software - Defined data Centre (SDDC):

A software Data Centre (SDDC) is a data centre in which computing, storage and networking resources are managed and controlled using software instead of relying mainly on physical Hardware

Architecture:



Working:

1) Physical resources such as servers, storage and networks are connected
2) Virtualization converts physical resources into virtual resource

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ii) Memory utilization:
Amount of RAM used to compare with total allocated memory

iii) Storage utilization:
Amount of allocated storage currently used

iv) Network utilization:
Amount of network bandwith consumed

$$\text{CPU utilization} = \frac{\text{CPU Time Used}}{\text{Total Available CPU Time}} \times 100$$

Working of VM utilization Analysis:

i) Collect VM performance Data
⇓
ii) Measure CPU, Storage and Network
⇓
iii) Analyse utilization Levels
⇓
iv) Identify overload or underutilized
⇓
v) optimized resource allocation

5) Secure identity and privacy Design in Multi-cloud federation.

Introduction:

A multi-cloud federation is a system in which multiple cloud services providers work together to share resources and services. Each provider may use different identity management systems, authentication methods, and access policies.

Analysis of the Issue:

i) Different identity formats: one cloud uses an email id, while another uses a user name or employee ID

ii) Authentication differences: Providers may use different login mechanisms.

